VCE Algorithmics

Unit 4, Week 2

Time Complexity Analysis

For each of the algorithms below, briefly explain what the algorithm does, and find the worst case time complexity, giving an explanation. (This can be in the form of annotations to the code.)

```
def list_sum(arr):
    total = 0
    for num in arr:
        total += num
    return total

def find_pair_with_sum(arr, target):
    n = len(arr)
    for i in range(n):
        for j in range(i + 1, n):
            if arr[i] + arr[j] == target:
                return True
    return False

def knapsack(array, target, limit):
# array contains pairs (weight, value)
   combs_to_try = empty list
   For each item in array:
      Duplicate combs_to_try
      Add item to the first half of combs_to_try
   max = 0
   For each line in combs_to_try:
      Sum weights in line
      If weights <= target:
         Sum values in line
         If values > max:
            maxline = line
   return maxline
```